

PROBLEM STATEMENT NO 1-ABC RETAIL MART: DECLINING SALES AND CUSTOMER SATISFICATION

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Course: MBA (Finance & Business Analytics)

Mapped Co1: Describe the role of Business Intelligence and Data Visualization

1.Business Problem Identification:

ABC Retail Mart is a retail chain operating through multiple branches, and in recent months, the company has been facing a drop in both sales and customer satisfaction. Although the branches collect a large amount of information related to sales, inventory, billing, and customer feedback, the data is maintained separately in different files and reports. Because of this, management does not have a single view of the company's overall performance. It also becomes difficult to understand why sales are decreasing and why customers are becoming less satisfied. The main challenge is not the lack of data, but the difficulty in bringing the available information together and using it effectively. This makes it harder for management to identify problems quickly and take appropriate business decisions.

2. Role of Business Intelligence in Addressing Problem:

Business Intelligence (BI) can help ABC Retail Mart bring together the information currently stored in different departments. Data from sales, inventory, billing, and customer feedback can be collected and stored in one central system. This makes it easier for managers to look at information from different areas at the same time instead of checking separate spreadsheets. The data can be cleaned, combined, and organized so that branch performance, product sales, and customer responses can be compared more easily. This analysis may help the company find issues such as products being out of stock, pricing problems, or poor customer service that could be affecting sales. Overall, BI helps turn the company's existing data into useful information that managers can use to understand the reasons for the sales decline and take better-informed decisions.

3. Role of Data Visualization in Decision-Making:

Business Intelligence helps in collecting and organizing the data, while data visualization helps managers understand the information more easily. This is especially useful for managers who may not have a technical background. A dashboard showing sales performance, stock movement, customer satisfaction, and billing information can give managers a quick view of what is happening in the business. For example, they can identify a branch where sales have suddenly fallen or notice an increase in complaints about a particular product. Charts such as line graphs, bar charts, and heat maps can make these changes easier to notice without going through large amounts of data manually. Therefore, data visualization helps connect the results of data analysis with actual business decisions and allows managers to respond to problems more quickly.

4. Business Functions Involved:

The decrease in sales and customer satisfaction is related to several areas of the business. Each department has information that can help in understanding the situation.

- Sales – Maintains information about revenue, number of transactions, and sales performance of individual branches.
- Inventory / Supply Chain – Monitors stock availability, stock levels, product movement, and replenishment activities.
- Customer Service – Collects customer feedback, complaints, reviews, and satisfaction survey information.
- Billing / Finance – Maintains transaction records, discounts, refunds, payment details, and billing-related information.
- Marketing – Handles promotional activities, pricing plans, customer engagement, and marketing campaigns.

5. Required Data Sources:

To understand the reasons behind the decline in sales and customer satisfaction, information from different parts of the organization should be brought together. The main sources of data include:

- Point-of-Sale (POS) Systems – Provides daily sales, transaction, and revenue details for each branch.
- Inventory Management System – Contains information about stock levels, product availability, reorder activity, and stock-out situations.
- Customer Relationship Management (CRM) System – Provides customer feedback, complaints, survey responses, and satisfaction-related information.
- Billing and Finance Records – Includes invoice details, refunds, payment methods, discounts, and billing errors.
- Loyalty Program / Membership Database – Helps identify repeat customers, purchase frequency, and customer retention patterns.
- Social media and Online Reviews – Provide additional information about customer opinions and their experiences with different branches or products.

6. Suggested Key Performance Indicators (KPIs):

The following KPIs can be monitored regularly to understand the performance of different business functions:

Business Function	Key Performance Indicator (KPI)	Purpose
Sales	Monthly and branch-wise sales growth rate	To find out which branches are contributing to the overall fall in sales
Customer Service	Customer Satisfaction Score (CSAT) / Net Promoter Score (NPS)	To understand how customer satisfaction is changing over time
Inventory	Stock-out rate and inventory turnover ratio	To identify whether product availability is affecting sales
Billing	Average transaction value and billing error rate	To identify possible problems with billing, pricing, or checkout
Marketing	Footfall-to-conversion rate and promotion redemption rate	To understand whether promotional activities are generating sales
Customer Retention	Repeat purchase rate and churn rate	To monitor customer loyalty and identify loss of regular customers

7. Data Collection Plan:

Before starting the detailed analysis and dashboard creation, ABC Retail Mart should organize the available data in a systematic manner. The following steps can be followed:

Step	Activity	Responsible / Source	Frequency
1	List and document the data sources available at each branch, including POS, CRM, inventory, and billing systems	IT Team / Branch Managers	One-time initial review
2	Bring data formats, column names, and reporting methods into a common standard across branches	IT Team / BI Team	Initially and whenever changes are required
3	Collect and combine the required data in a central data warehouse	BI Analyst / IT Team	Daily or weekly
4	Check the data for missing values, duplicate records, and other errors	BI Analyst	Weekly
5	Develop dashboards using the selected KPIs	BI Analyst	Initial development and regular updates
6	Present the dashboards to management and collect their suggestions	Management Team	Monthly
7	Review the KPIs and update the data requirements when new business issues are identified	BI Analyst / Management	Quarterly

8. Conclusion:

ABC Retail Mart already has information from several areas of its business, but the data is maintained separately, making it difficult to understand the complete situation. By bringing together sales, inventory, billing, and customer feedback data through a Business Intelligence system, the company can get a clearer view of its performance. Interactive dashboards can then present this information in a simple form and help management identify branches, products, or business processes that may be contributing to the decline. Regularly monitoring the selected KPIs and following a proper data collection process can provide useful information for management. This can support quicker responses to business problems and help the company make decisions based on actual data rather than assumptions.